

Elaborated Claude Slide Deck (PPTX)

Prompt

Project: Automated: A Multi-Agentic Trading System

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Created: May 2026



Operational Instructions:

Copy the entire prompt block below and paste it directly into Claude. Claude will act as a Presentation Designer and produce a fully functional, premium Python script using `python-pptx` to programmatically output a gorgeous, 10-slide PowerPoint (.pptx) file.

THE PROMPT TO COPY & PASTE TO CLAUDE:

You are an Elite Presentation Designer, UI/UX Architect, and Chief Technical Explainer. Your goal is to design an outstanding, highly professional, 10-slide PowerPoint presentation (.pptx) for our final BSCS thesis defense: "Automated: A Multi-Agentic Trading System".

To make this presentation fully tangible, you must write a complete, executable Python script using the ``python-pptx`` library that programmatically compiles all 10 slides with a premium corporate dark-mode theme, pixel-perfect layout shapes, structured tables, text boxes, and full speaker notes embedded in the metadata of every slide.

🍷 PRESENTATION VISUAL STANDARDS (Bloomberg-Inspired Theme)

* **Slide Dimensions**: Widescreen 16:9 layout.

* **Color Palette**:

* Background: Pitch Dark Slate (`#0B0F17`)

* Container Cards: Slate Charcoal (`#131C2E`)

* Primaries & Accent Outlines: Bright Icy Cyan (`#38BDF8`)

* Strategic BUY Indicators: Neon Emerald (`#10B981`)

* Strategic SELL/Alert Vectors: Vivid Orange-Red (`#F97316`)

* Body Text: Clean Light Silver (`#E2E8F0`)

* **Typography**: Modern Sans-Serif (Arial or Segoe UI) with highly disciplined font-size hierarchies.

📊 SLIDE-BY-SLIDE CONTENT & LAYOUT STRUCTURE

📄 Slide 1: Title Slide & Thesis Overview

* **Layout**: Centered premium typography on dark background with horizontal icy-cyan accent line.

* **Content**:

* Title: "Automated: A Multi-Agentive Trading System"

* Subtitle: "An Asynchronous Decoupled Multi-Agent Portal with Neon PostgreSQL and Langfuse Telemetry"

* Presenters: Abdul Qadeer Khan & Ahmad Abdullah

* Thesis details: Bachelor of Computer Science Final Year Project

📄 Slide 2: The Core Problem & Architecture Paradigm

* **Layout**: Two-column visual layout comparison.

* **Content**:

* Left Column: "The Monolith Bottleneck" (context window congestion, single-prompt contamination, unreliability).

* Right Column: "The Decoupled Paradigm" (decoupled agents, absolute sandboxing, immutable state channels, predictable overrides).

📄 Slide 3: Multi-Agent Decoupled Orchestration Flow

* **Layout**: Graphic layout flow depicting agent registry connections.

* **Content**:

* Step 1: Master Desk Router (Entry checkpoint & strict ticker parameter controls).

* Step 2: Intraday & Swing Market Analyst (Calculates technical matrices, returns RSI/EMA, zero execution authority).

* Step 3: Risk Manager Agent (Compliance auditor, enforces strict APPROVED/REJECTED verdicts with math).

📄 Slide 4: Dynamic Strategy Parameters & Data Matrix

* **Layout**: A beautifully structured table spanning 4 core execution strategies.

* **Content**:

* Rows: Intraday Scalp (15m), Intraday Swing (1h), Medium Swing (4h), Macro Position (1d).

* Columns: Lookback Period, Indicators, Target Exit Ratios (Stop Loss / Take Profit bounds).

📄 Slide 5: Secure Operator Gateway & Database Registry

* **Layout**: Right card showing user model table fields, left showing validation

rules.

* **Content**:

- * Neon PostgreSQL `system\_users` schema (email unique indexing, bcrypt hashing).

- * Strict validation criteria: cellphone regex (`^(03|\+923)\d{9}\$`), name trimming, and the optimized 6-character flexible password accessibility rule.

📺 Slide 6: The Transaction-Isolated Startup Migration Fix

* **Layout**:

- Logical split-column detailing the startup fix.

* **Content**:

- * The Problem: Aborted transaction warnings due to failing rename column commands.

- * The Solution: Splitting startup migrations into two separate connection blocks (Block 1: RENAME column with explicit transaction rollback; Block 2: Schema `create\_all` database creation).

📺 Slide 7: Interactive SVG Candlestick Charting Engine

* **Layout**:

- Horizontal wireframe description representing SVG canvas blocks.

* **Content**:

- * Real-time yfinance fetching & local min/max price range scaling.

- * Adjusting SVG coordinate inversion (Y=0 at top) with 8% padding.

- * Plotting beating wicks and body blocks with 10-second polling updates.

📺 Slide 8: Telemetry, Latency Tracing & Observability

* **Layout**:

- Graphic panels representing Langfuse monitoring blocks.

* **Content**:

- * Global trace initialization at user entry.

- * Nested execution spans to audit individual agent token usage and latency.

- * Database insert payload checks to monitor data pipelines.

📺 Slide 9: Cloud Deployment & Render Best Practices

- * **Layout**: Three-step horizontal deployment timeline cards.

- * **Content**:

- * Build command execution (`pip install -r requirements.txt`).

- * Web Service starting configurations (`uvicorn main:app --host 0.0.0.0 --port \$PORT`).

- * Handling Render Free-tier container spin-down inactivity cold starts.

📺 Slide 10: Conclusion & Core Contributions

- * **Layout**: Clean central focal card summarizing final results.

- * **Content**:

- * Successful Neon PostgreSQL persistence and agent decision pipeline execution.

- * Academic contributions: Multi-Agent decoupling and automated risk-hedging guardrails.

- * Prominent open section: "Questions & Thesis Defense Q&A".

🖥️ PYTHON-PPTX PROGRAMMATIC CODE GENERATION REQUIREMENTS

Your final python script must:

1. Define a pitch-black dark theme colors using `RGBColor()`.
2. Add text boxes programmatically with custom left/top/width/height measurements to avoid element overlaps.
3. Apply correct font size formatting (Titles: 28-36pt, Subtitles: 18pt, Bullets: 14-16pt).

4. Embed the precise **Speaker Notes** inside each slide's
`.notes\_slide.notes\_text\_frame` so we have perfect scripts to read during the presentation!
5. Save the generated file as `automated\_trading\_presentation.pptx`.